

Absence of Menstrual Periods

(Amenorrhea)

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Having no menstrual periods is called amenorrhea.

Amenorrhea is normal in the following circumstances:

- Before puberty
- During pregnancy
- While breastfeeding
- After menopause

At other times, it may be the first symptom of a serious disorder.

Amenorrhea may be accompanied by other symptoms, depending on the cause. For example, women may develop masculine characteristics (virilization), such as excess body hair ([hirsutism](#)), a deepened voice, and increased muscle size. They may have headaches, vision problems, or a decreased sex drive. They may have difficulty becoming pregnant.

In most women with amenorrhea, the ovaries do not release an egg. Such women cannot become pregnant.

If amenorrhea lasts a long time, problems similar to those associated with [menopause](#) may develop. They include hot flashes, vaginal dryness, decreased bone density ([osteoporosis](#)), and an increased risk of heart and blood vessel disorders. Such problems occur because in women who have amenorrhea, the estrogen level is low.

Types of amenorrhea

There are 2 main types of amenorrhea:

- **Primary:** Menstrual periods never start.

- **Secondary:** Periods start, then stop.

Usually if periods never start, girls do not go through puberty, and thus secondary sexual characteristics, such as breasts and pubic hair, do not develop normally.

If women have been having menstrual periods, which then stop, they may have secondary amenorrhea. Secondary amenorrhea is much more common than primary.

Hormones and menstruation

Menstrual periods are regulated by [a complex hormonal system](#). Each month, this system produces hormones in a certain sequence to prepare the body, particularly the uterus, for pregnancy. When this system works normally and there is no pregnancy, the sequence ends with the uterus shedding its lining, producing a [menstrual period](#). The hormones in this system are produced by the following:

- Hypothalamus (part of the brain that helps control the pituitary gland)
- Pituitary gland, which produces luteinizing hormone and follicle-stimulating hormone
- Ovaries, which produce estrogen and progesterone

Other hormones, such as thyroid hormones and prolactin (produced by the pituitary gland), can affect the menstrual cycle.

Causes of Amenorrhea

Conditions that can cause amenorrhea include hormonal disorders, birth defects, genetic disorders, medications, and illicit drugs.

The most common reason for amenorrhea in women who are not pregnant or breastfeeding is malfunction of any part of the hormonal system (hypothalamus, pituitary gland, and ovaries). When this system malfunctions, the ovaries do not release an egg. The type of amenorrhea that results is called ovulatory dysfunction.

Amenorrhea can also result from conditions that affect the uterus, cervix, or vagina.

Less commonly, the hormonal system is functioning normally, but another problem prevents periods from occurring. For example, menstrual bleeding may not occur because the uterus is scarred (Asherman syndrome), because the cervix is narrowed (cervical stenosis), or because a birth defect blocks the flow of menstrual blood out of the vagina.

Which causes are most common depends on whether amenorrhea is primary or secondary.

Primary amenorrhea

The disorders that cause primary amenorrhea are relatively uncommon, but the most common are a:

- Genetic disorder
- Birth defect of the reproductive organs that blocks the flow of menstrual blood (such as an [imperforate hymen](#))

Genetic disorders include:

- [Turner syndrome](#): absence of all or part of 1 of the 2 X chromosomes
- Kallmann syndrome: caused by a decrease in female sex hormones
- Overproduction of male hormones by the adrenal glands (congenital adrenal hyperplasia)
- [Genital disorders](#) that result in ambiguous—neither male nor female—genitals (pseudohermaphroditism or true hermaphroditism)
- Disorders that result in having a Y chromosome (which normally occurs only in males).

Genetic disorders and birth defects that cause primary amenorrhea may not be noticed until puberty. These disorders cause only primary amenorrhea, not secondary.

Sometimes puberty is delayed in girls who do not have a disorder, and normal periods simply begin at a later age. Such [delayed puberty](#) may run in families.

Secondary amenorrhea

The **most common causes** are:

- Pregnancy
- Breastfeeding
- Malfunction of the hypothalamus
- [Polyendocrine metabolic ovarian syndrome](#)
- [Premature menopause](#) (primary ovarian insufficiency)
- Malfunction of the pituitary gland or the thyroid gland
- Use of certain medications, such as birth control pills (oral contraceptives), antidepressants, or antipsychotic medications

Pregnancy is the most common cause of amenorrhea among women in the reproductive years.

The **hypothalamus** may malfunction for several reasons, including the following:

- Inadequate calorie intake due to dieting, orthorexia (obsession with eating healthy foods), eating disorders (such as [anorexia nervosa](#), [bulimia](#)), [undernutrition](#) (for example, due to food insecurity), or [malabsorption](#)
- Excessive exercise (as done by competitive athletes, particularly women who participate in sports that involve maintaining a low body weight)
- Metabolic disturbances (for example, those associated with obesity or aging)
- Emotional stress
- Mental disorders (such as [depression](#) or [obsessive-compulsive disorder](#))
- Some medications, including psychoactive medications (eg, antipsychotics, antidepressants) and antiepileptics
- Misuse of some substances (eg, alcohol, opioids)

The **pituitary gland** may malfunction because:

- It is damaged by a disorder (such as a tumor) or a head injury.
- Levels of prolactin are high.

Antidepressants, antipsychotic medications, oral contraceptives (sometimes), or certain other medications can cause prolactin levels to increase, as can pituitary tumors and some other disorders.



Medications That Can Cause Menstrual Periods to Stop

Type	Examples	Symptoms
Medications and drugs that can increase the production of prolactin*		
Antihypertensives	<u>Methyldopa</u>	Production of breast milk in women who are not breastfeeding
	Reserpine	
	<u>Verapamil</u>	
Antipsychotics	<u>Haloperidol</u>	
	<u>Molindone</u>	
	<u>Olanzapine</u>	
	Phenothiazines	
	<u>Pimozide</u>	
	<u>Risperidone</u>	
Illicit drugs	<u>Cocaine</u>	
Estrogens	—	
Medications used to treat digestive disorders	<u>Cimetidine</u>	
	<u>Metoclopramide</u>	
Opioids	<u>Codeine</u>	
	<u>Morphine</u>	
Antidepressants selective serotonin reuptake inhibitors, (tricyclics, and monoamine oxidase inhibitors)	<u>Fluoxetine</u>	
	<u>Clomipramine</u>	
	<u>Phenelzine</u>	
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Medications that affect the balance of female and male hormones		
Synthetic androgens	<u>Danazol</u>	Development of masculine characteristics (such as excess body hair, a



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